Both endpoint asymptotics of the first zero of $E_{a,a}(-t^a)$

Principia Math In Progress

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Task 1: MathDB 361624. Put

$$H(a,t) = E_{a,a}(-t^a) = \sum_{k\ge 0} \frac{(-1)^k t^{ak}}{\Gamma(a(k+1))}, \qquad Z(a) = \min\{t > 0 : H(a,t) = 0\}.$$

Conjecture 1(b) of Chaudhary, Diethelm and Hashemishahraki [1] proposes $Z(a) = (c\log(a-1) + d)(1+o(1))$ as $a \downarrow 1$, with $c \approx -0.81$ and $d \approx 2.25$. Part (c) proposes $Z(a) = \pi + c(2-a) + o(2-a)$ as $a \uparrow 2$, again with $c \approx -0.81$. The authors explicitly ask whether these two coefficients are equal. The following theorem answers this question and corrects the first coefficient.

Theorem 1. *The first positive zero exists for all $a > 1$ sufficiently close to either endpoint 1 or 2. With $L = \log(1/\varepsilon)$,*

$$\boxed{Z(1+\varepsilon) = L + 2\log L + o(1)} \qquad (\varepsilon \downarrow 0),$$

$$\boxed{Z(2-\delta) = \pi + c_*\delta + O(\delta^2), \quad c_* = \frac{\pi\,\mathrm{Ci}(\pi) - \mathrm{Si}(\pi)}{2}} \qquad (\delta \downarrow 0).$$

Here $-811/1000 < c_* < -809/1000$, so the two leading coefficients are unequal: the first is -1, and the second is $-0.8102512403460584\ldots$.

We use $\mathrm{Si}(x) = \int_0^x \sin u/u\,du$ and $\mathrm{Ci}(x) = \gamma + \log x + \int_0^x (\cos u - 1)/u\,du$ for $x > 0$. The theorem refutes the numerical leading coefficient in part (b) and hence the conjunction recorded in MathDB. It also supplies the exact coefficient in part (c). The global positivity bound $H(a,t) > 0$ for $0 \le t \le 2.93785$ in part (a) remains unresolved. No contradiction rests on the constant d, which is invisible at the source's multiplicative $1 + o(1)$ precision.

An exact representation near order one

Write $a = 1 + \varepsilon$, $0 < \varepsilon < 1/4$, $\theta = \pi\varepsilon$, and

$$K(r) = \frac{\sin\theta}{\pi} \frac{r^{a-1}}{(r^a - 1)^2 + 2r^a(1 - \cos\theta)} \quad (r > 0).$$

For $f_a(t) = E_{a,1}(-t^a)$ the classical residue formula is

$$f_a(t) = \frac{2}{a} e^{t\cos(\pi/a)} \cos(t\sin(\pi/a)) - \int_0^\infty e^{-tr} K(r)\,dr. \tag{1}$$

Indeed, termwise Laplace transformation for real $s > 1$ gives $s^{a-1}/(s^a+1)$. On its principal branch the poles are $e^{\pm i\pi/a}$, with residues in the inverse transform $a^{-1}e^{te^{\pm i\pi/a}}$. The imaginary part of the transform on the upper side of the negative real cut is

$$\operatorname{Im} \frac{r^{a-1}e^{i\theta}}{1-r^a e^{i\theta}} = \pi K(r).$$

Deforming the inversion contour gives the negative integral in (1). The small-circle contribution is $O(\rho^a)$ and the large arcs vanish for $t > 0$, using the transform's $O(1/s)$ decay and the exponential on the left arcs. The cut integral converges absolutely. These representation methods are classical; see [2, 3].

Differentiating the defining series gives $-f_a'(t) = t^{a-1}H(a,t)$. Differentiating the exact integral in (1) is justified for $t > 0$ by exponential integrability. Thus, with $g_a(t) = t^{a-1}H(a,t)$,

$$g_a(t) = -\frac{2}{a}e^{t\cos(\pi/a)}\cos\bigl(t\sin(\pi/a)+\pi/a\bigr) - \int_0^\infty re^{-tr}K(r)\,dr. \tag{2}$$

This is differentiation of an exact identity, not of an asymptotic expansion. The unweighted mass, by $u = r^a$, is

$$\int_0^\infty K(r)\,dr = \frac{\sin\theta}{\pi a}\int_0^\infty \frac{du}{(u-\cos\theta)^2+\sin^2\theta} = \frac{\pi-\theta}{\pi a} = \frac{2}{a} - 1. \tag{3}$$

Uniform estimates and the first zero near order one

We prove, uniformly for $L/4 \le t \le 3L/2$,

$$g_a(t) = e^{-t}(1+o(1)) - \varepsilon\Gamma(a+1)t^{-a-1}(1+o(1)), \tag{4}$$

with separate relative errors for the two displayed terms. All constants below are uniform for sufficiently small ε. Write $D(r) = 1 - 2r^a\cos\theta + r^{2a}$ and $h = t^{-2}$. For $r \le 1/2$, $D \ge 1/4$ and $D^{-1} = 1 + O(r^a)$, so

$$\begin{aligned}\int_0^{1/2} re^{-tr}K(r)\,dr &= \frac{\sin\theta}{\pi}\Bigl\{\Gamma(a+1)t^{-a-1} + O(t^{-2a-1}) + O\bigl(t^{-a-1}(1+t)^2e^{-t/2}\bigr)\Bigr\}\\ &= \varepsilon\Gamma(a+1)t^{-a-1}(1+o(1)).\end{aligned}$$

The algebraic error follows by extending its gamma integral to infinity. The omitted main-integral tail follows by $u = tr$ and $u^a \le 1 + u^2$, since $1 < a < 5/4$.

Outside $[1/2, 3/2]$ the unweighted K mass is $O(\varepsilon)$: on the lower interval use $D \ge 1/4$; on the upper interval use $D \ge (r^a-1)^2 \ge r^{2a}/9$, whence $K(r) \le C\varepsilon r^{-a-1}$. On $[1/2, 3/2]$ the mean value theorem yields $|r^a - 1| \ge C_0|r-1|$, so $K(r) \le C\varepsilon/(r-1)^2$ for $r \ne 1$. Equation (3) therefore implies

$$\int_{1-h}^{1+h} K(r)\,dr = 1 + O(\varepsilon) + O(\varepsilon/h) = 1 + o(1).$$

Here $\varepsilon t^2 \to 0$. In this central interval, $re^{-tr} = e^{-t}(1+O(h+th))$, and consequently its contribution to the integral in (2) is $e^{-t}(1+o(1))$. The rest of $[1/2, 3/2]$ contributes at most $C\varepsilon t^4 e^{-t/2} = o(e^{-t})$, since its ratio to e^{-t} is at most $CL^4\varepsilon^{1/4}$. For the remaining upper tail we retain the exponential weight:

$$\int_{3/2}^\infty re^{-tr}K(r)\,dr \le C\varepsilon\int_{3/2}^\infty r^{-a}e^{-tr}\,dr \le C\varepsilon e^{-3t/2}/t = o(e^{-t}).$$

Finally $\cos(\pi/a) = -1 + O(\varepsilon^2)$ and $\sin(\pi/a) = O(\varepsilon)$. The residue term in (2) is $2e^{-t}(1 + o(1))$ because $\varepsilon t \to 0$ uniformly. Subtracting the central and small-r contributions proves (4).

We next exclude earlier zeros. For $1 \le b \le a$ and $k \ge 1$,

$$\partial_b \frac{t^{bk}}{\Gamma(b(k+1))} = \frac{t^{bk}}{\Gamma(b(k+1))}\big(k \log t - (k+1)\psi(b(k+1))\big).$$

We have $\Gamma(b(k+1)) \ge k!$ and $0 < \psi(b(k+1)) \le \log(b(k+1)) \le 2(k+1)$. The $k = 0$ derivative is uniformly bounded. For $1 \le t \le L/4$, eventually $t^{b-1} \le 2$, and summing the factorial majorant gives $|\partial_b H(b,t)| \le C(1+t)^3 e^{2t}$. As $H(1,t) = e^{-t}$, integration over b yields

$$\frac{|H(a,t) - e^{-t}|}{e^{-t}} \le C\varepsilon(1+L)^3 e^{3L/4} = o(1).$$

For $0 < t \le 1$, use $kt^{bk}|\log t| \le 1/(b\mathrm{e})$; the derivative series is uniformly bounded by a summable multiple of $(1+(k+1)^2)/k!$. Hence $H(a,t) = e^{-t} + O(\varepsilon)$ uniformly there. Also $H(a,0) = 1/\Gamma(a) > 0$. These bounds justify the differentiations and show positivity on $[0, L/4]$ for small ε.

Fix $0 < \eta < 1/2$. The ratio of the second leading term in (4) to the first is

$$Q_\varepsilon(t) = \varepsilon \Gamma(a+1) e^t t^{-a-1}.$$

It is increasing for $t > a + 1$. At $(1-\eta)L$ it tends to zero, and at $(1+\eta)L$ it tends to infinity. Equation (4) therefore gives positivity throughout $[L/4, (1-\eta)L]$ and a negative value at $(1+\eta)L$. Since g and H have the same sign for $t > 0$, continuity and earlier positivity show that a first positive zero exists and $Z(a)/L \to 1$. At this zero the two relative errors in (4) tend to zero. Taking logarithms gives

$$Z(a) = L + (a+1)\log Z(a) - \log\Gamma(a+1) + o(1) = L + 2\log L + o(1).$$

The last step uses $\log Z(a) = \log L + o(1)$, $\varepsilon \log Z(a) = o(1)$, and $\log\Gamma(a+1) = o(1)$.

The coefficient near order two

Set $F(a,t) = tH(a,t)$. Then $F(2,t) = \sin t$ and $F_t(2,\pi) = -1$. The series and its first two derivatives converge locally uniformly near $(2,\pi)$: for $a \ge 3/2$ and $k \ge 1$, $\Gamma(a(k+1)) \ge k!$, and derivatives contribute only polynomial and logarithmic factors in k. The $k = 0$ term is smooth. The implicit function theorem gives a C^2 zero branch $z(a)$, and its coefficient in $2 - a$ is $-F_a(2,\pi)$. For small fixed $\eta > 0$, $F_t(a,t) < -1/4$ near $[\pi - \eta, \pi + \eta]$ when a is close to 2. On $[0, \pi - \eta]$, $H(a,t)$ converges uniformly to $\sin t/t$ (value 1 at zero), which has a positive minimum. Thus the branch is the first positive zero.

To evaluate its derivative, introduce $F_2(a,t) = tE_{a,2}(-t^a)$ and $w(t) = \partial_a F_2(2,t)$. Termwise differentiation and $\psi(m+1) = H_m - \gamma$ give

$$w(t) = \sum_{k\ge1} \frac{(-1)^k k t^{2k+1}}{(2k+1)!}(\log t + \gamma - H_{2k+1}) = -\frac{1}{2}\int_0^t (\log(t-u) + \gamma) u \sin u \, du.$$

The integral identity follows by expanding $u \sin u$ and using $\int_0^t u^m \log(t-u)\, du = t^{m+1}(\log t - H_{m+1})/(m+1)$. With $v = \pi - u$, two integrations by parts yield

$$-w(\pi) = \frac{1}{2}\int_0^\pi (\log v + \gamma)(\pi - v)\sin v \, dv = \frac{\pi\,\mathrm{Ci}(\pi) + \mathrm{Si}(\pi)}{2}.$$

For explicit primitives, use $-\log v\cos v+\mathrm{Ci}(v)$ for $\log v\sin v$ and $-v\log v\cos v+(\log v+1)\sin v-\mathrm{Si}(v)$ for $v\log v\sin v$. Their limits at zero are γ and 0.

The derivative of the second parameter in F contributes one extra digamma factor. Thus the required coefficient is

$$\begin{aligned} c_* &= -w(\pi)+\sum_{k\ge 0}\frac{(-1)^k\pi^{2k+1}\psi(2k+2)}{(2k+1)!} \\ &= -w(\pi)+\int_0^1\frac{\sin\pi-\sin(\pi u)}{1-u}\,du = -w(\pi)-\mathrm{Si}(\pi). \end{aligned}$$

Here the Euler-constant part vanishes, and $H_{2k+1}=\int_0^1(1-u^{2k+1})/(1-u)\,du$ justifies the middle equality by absolute convergence of the integrated series. This proves the closed formula; Taylor's theorem gives the stated $O(\delta^2)$ remainder.

An exact rational enclosure

We specify a finite rational certificate for $c_*\neq -1$. Let $p_-=31415926535/10^{10}$ and $p_+=31415926536/10^{10}$. Machin's identity $\pi=16\arctan(1/5)-4\arctan(1/239)$, with 12 alternating terms and the next-term bound for each arctangent, proves $p_-<\pi<p_+$. For rational $x>1$ put $y=(x-1)/(x+1)$ and

$$A(x)=2\sum_{j=0}^{49}\frac{y^{2j+1}}{2j+1},\qquad B(x)=A(x)+\frac{2y^{101}}{101(1-y^2)}.$$

Then $A(x)<\log x<B(x)$. With $N=2048=2^{11}$,

$$H_N-11B(2)-1/N<\gamma<H_N-11A(2),\qquad A(p_-)<\log\pi<B(p_+).$$

The gamma bound follows from $0<H_N-\log N-\gamma<1/N$, obtained by summing $0<\log(1+1/j)-1/(j+1)<1/j-1/(j+1)$ for $j\ge N$. Use the alternating series

$$\begin{aligned} \mathrm{Ci}(\pi) &= \gamma+\log\pi+\sum_{k=1}^{10}\frac{(-1)^k\pi^{2k}}{2k(2k)!}+R_C, & -\frac{p_+^{22}}{22\cdot 22!}<R_C<0, \\ \mathrm{Si}(\pi) &= \sum_{k=0}^{10}\frac{(-1)^k\pi^{2k+1}}{(2k+1)(2k+1)!}+R_S, & -\frac{p_+^{23}}{23\cdot 23!}<R_S<0. \end{aligned}$$

The term magnitudes decrease, as follows from their ratios and $\pi^2<10$. Evaluate positive powers in $[p_-^j,p_+^j]$, add intervals with their displayed signs, and use all four endpoint products when multiplying intervals. The exact resulting rational endpoints c_-,c_+, stored by the verifier, satisfy

$$-811/1000<c_-<c_*<c_+<-809/1000.$$

Together with the first asymptotic, this proves the coefficients unequal and completes Theorem 1. □

Verification, attribution, and scope

The representation and perturbation methods are classical. The uniform weighted-kernel estimates above and the first-zero arguments are included explicitly. Companion MathDB 361626 treats second parameter 1 near order 1; MathDB 361625 treats fixed second parameter 2 near order 2. Here the second parameter varies with the order, changing both coefficients. These are distinct source conjectures. Earlier first-zero existence and monotonicity results include [4]; no novelty for those general facts is claimed.

The verifier checks rational constants, the coefficient enclosure, and symbolic identities exactly. Supplemental 110-digit calculations compare the exact integral with the series and examine both endpoint limits. These are not interval certificates or a substitute for first-zero minimality. The complete current primary and author-posted target-final text were read; limitations of the bounded priority audit are recorded separately. Review is by the authoring agent only.

Task 2: Determine the next term after $L + 2\log L$, with an error bound.

Task 3: Prove or refute the global positivity bound in Conjecture 1(a).

Task 4: Determine the quadratic coefficient of $(2-a)^2$ in $Z(a)$.

Task 5: Prove uniqueness and locate the global minimum of $Z(a)$ on $(1, 2]$ with rigorous bounds.

References

[1] R. Chaudhary, K. Diethelm and S. Hashemishahraki, On the separation of solutions to fractional differential equations of order $\alpha \in (1, 2)$, *Applied Numerical Mathematics* **203** (2024), 84–96, doi:10.1016/j.apnum.2024.05.020. Current preprint: arXiv:2401.14771v3. Conjecture 1 in both versions.

[2] J. W. Hanneken, B. N. N. Achar and D. M. Vaught, An alpha-beta phase diagram representation of the zeros and properties of the Mittag-Leffler function, *Advances in Mathematical Physics* (2013), Article 421685, doi:10.1155/2013/421685.

[3] R. B. Paris, Exponential asymptotics of the Mittag–Leffler function, *Proceedings of the Royal Society A* **458** (2002), 3041–3052, doi:10.1098/rspa.2002.0977.

[4] P. W. Eloe and Y. Li, Existence, comparison, and monotonicity of principal eigenvalues of fractional Sturm–Liouville problems involving Riemann–Liouville derivatives, *Fractional Calculus and Applied Analysis* **29** (2026), 1–25, doi:10.1007/s13540-026-00481-1.